

FOURIER SERIES AND PDES

Fourier Series:

1. Find the period of $f(x) = \sin \frac{n\pi x}{L}$

Here, $f(x) = \sin \frac{n\pi x}{L}$

Let T be the period of $f(x)$. Then,

$$f(x+T) = f(x)$$

$$\therefore \sin \frac{n\pi}{L} (x+T) = \sin \frac{n\pi x}{L}$$

$$\therefore \sin \left(\frac{n\pi x}{L} + \frac{n\pi T}{L} \right) = \sin \frac{n\pi x}{L}$$

$$\therefore \sin \frac{n\pi x}{L} \cos \frac{n\pi T}{L} + \cos \frac{n\pi x}{L} \sin \frac{n\pi T}{L} = \sin \frac{n\pi x}{L}$$

Equating the coefficient of $\sin \frac{n\pi x}{L}$ on both sides,

$$\cos \frac{n\pi T}{L} = 1 = \cos 2\pi$$

$$\therefore \frac{n\pi T}{L} = 2\pi$$

$$\therefore T = \underline{\underline{\frac{2L}{n}}}$$

2. Find the Fourier series of the following functions:

(i) $f(x) = x$; $-\pi < x < \pi$

Let the Fourier series of given function $f(x) = x$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (i)}$$

$$\text{where, } a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx$$

$$= 0$$

[$\because f(x)$ is an odd function]

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos nx \, dx$$

$$= 0$$

[$\because x \cos nx$ is an odd function]

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin nx \, dx$$

$$= \frac{2}{\pi} \left[-\frac{x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \times \left(-\frac{\pi \cos n\pi}{n} \right)$$

$$= -\frac{2}{n} (-1)^n \quad [\because \cos n\pi = (-1)^n]$$

So, from (i),

$$f(x) = 0 + \sum_{n=1}^{\infty} \left[0 + \left(-\frac{2}{n} \right) (-1)^n \sin nx \right]$$

$$= -2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin nx$$

(ii) $f(x) = x^2$; $-\pi < x < \pi$ and deduce the relations:

$$(a) \quad \frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} \dots = \sum \frac{(-1)^{n+1}}{n^2}$$

$$(b) \quad \frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} \dots = \sum \frac{1}{n^2}$$

$$(c) \quad \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} \dots = \sum \frac{1}{(2n-1)^2}$$

Let the Fourier series of given function $f(x) = x^2$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (1)}$$

$$\begin{aligned}
 \text{where, } a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx \\
 &= \frac{2}{2\pi} \int_0^{\pi} x^2 dx \\
 &= \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} \\
 &= \frac{1}{\pi} \times \frac{\pi^3}{3} \\
 &= \frac{\pi^2}{3}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx \\
 &= \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx \\
 &= \frac{2}{\pi} \left[\cancel{\frac{x^2 \sin nx}{n}} + \frac{2x \cos nx}{n^2} - \cancel{\frac{2 \sin nx}{n^3}} \right]_0^{\pi} \\
 &= \frac{2}{\pi} \times \frac{2\pi}{n^2} \cos n\pi \\
 &= \frac{4}{n^2} (-1)^n
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx \\
 &= 0 \quad [\because x^2 \sin nx \text{ is an odd function}]
 \end{aligned}$$

So, from (i)

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx + 0 \right)$$

$$\text{i.e. } x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx \quad \text{--- (ii)}$$

(a) Put $x=0$ in (ii), we get

$$0 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos 0$$

$$\therefore \frac{\pi^2}{3} = -4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$\begin{aligned} \hookrightarrow \frac{\pi^2}{12} &= - \left[-\frac{1}{1} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots \right] \\ \hookrightarrow \frac{\pi^2}{12} &= 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \quad \text{proved} \\ &\quad \text{--- (iii)} \end{aligned}$$

(b) Put $x = \pi$ in (ii), we get

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos n\pi$$

$$\hookrightarrow \pi^2 - \frac{\pi^2}{3} = 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} (-1)^n$$

$$\hookrightarrow \frac{2\pi^2}{3 \times 4} = \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n^2}$$

$$\begin{aligned} \hookrightarrow \frac{\pi^2}{6} &= \frac{1}{1} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \\ &= \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \text{proved} \\ &\quad \text{--- (iv)} \end{aligned}$$

(c) Adding (iii) and (iv), we get

$$\frac{\pi^2}{12} + \frac{\pi^2}{6} = 2 + \frac{2}{3^2} + \frac{2}{5^2} + \dots$$

$$\hookrightarrow \frac{\pi^2}{4} = 2 \left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$$

$$\begin{aligned} \hookrightarrow \frac{\pi^2}{8} &= 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \\ &= \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \quad \text{proved} \end{aligned}$$

(iii) $f(x) = x + |x|$; $-\pi < x < \pi$

Let the Fourier series of given function $f(x) = x + |x|$ with period $2l = 2\pi$ (i.e. $l = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (i)}$$

$$\text{Here, } |x| = \begin{cases} -x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$

$$\text{So, we have, } f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ 2x & \text{if } 0 \leq x < \pi \end{cases}$$

Now,

$$\begin{aligned} a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{2\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{2\pi} \int_0^{\pi} f(x) dx \\ &= \frac{1}{2\pi} \int_0^{\pi} 2x dx \\ &= \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} \\ &= \frac{1}{\pi} \times \frac{\pi^2}{2} \\ &= \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \int_0^{\pi} 2x \cos nx dx \\ &= \frac{2}{\pi} \left[\cancel{x \sin nx}^0 + \frac{\cos nx}{n^2} \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2} \right] \end{aligned}$$

$$= \frac{2}{\pi n^2} [(1-1)^n - 1]$$

$$= \begin{cases} -\frac{4}{\pi n^2} & \text{for odd } n \\ 0 & \text{for even } n \end{cases}$$

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 f(x) \sin nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx dx \\ &= \frac{1}{\pi} \int_0^{\pi} 2x \sin nx dx \\ &= \frac{2}{\pi} \left[\cancel{-x \cos nx}^0 + \frac{\sin nx}{n^2} \right]_0^{\pi} \end{aligned}$$

$$= \frac{2}{\pi} \times (-\pi \cos n\pi)$$

$$= -\frac{2}{n} (-1)^n$$

So, from (i)

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left[\frac{2}{\pi n^2} \cdot [(-1)^n - 1] \cos nx - \frac{2}{n} (-1)^n \sin nx \right]$$

$$= \frac{\pi}{2} + 2 \sum_{n=1}^{\infty} \left[\frac{n^2}{\pi} \{(-1)^n - 1\} \cos nx - \frac{(-1)^n}{n} \sin nx \right]$$

(iv) $f(x) = x \sin x$; $-\pi < x < \pi$

Let the Fourier series of given function $f(x) = x \sin x$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (1)}$$

where, $a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} x \sin x dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin x dx$$

$$= \frac{1}{\pi} \left[-x \cos x + \sin x \right]_0^{\pi}$$

$$= \frac{1}{\pi} \times (-\pi \cos \pi)$$

$$= 1$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin x \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin x \cos nx dx$$

$$\begin{aligned}
&= \frac{1}{\pi} \int_0^{\pi} x 2 \sin x \cos nx \, dx \\
&= \frac{1}{\pi} \int_0^{\pi} x [\sin(nx+x) - \sin(nx-x)] \, dx \\
&= \frac{1}{\pi} \int_0^{\pi} x \sin(nx+x) \, dx - \frac{1}{\pi} \int_0^{\pi} x \sin(nx-x) \, dx \\
&= \frac{1}{\pi} \left[\frac{-x \cos(n+1)x}{n+1} + \frac{\sin(n+1)x}{(n+1)^2} \right]_0^{\pi} - \frac{1}{\pi} \left[\frac{-x \cos(n-1)x}{n-1} + \frac{\sin(n-1)x}{(n-1)^2} \right]_0^{\pi} \\
&= \frac{1}{\pi} \left[\frac{-\pi \cos(n+1)\pi}{n+1} + 0 \right] - \frac{1}{\pi} \left[\frac{-\pi \cos(n-1)\pi}{n-1} + 0 \right] \\
&= -\frac{(-1)^{n+1}}{n+1} + \frac{(-1)^{n-1}}{n-1} \\
&= (-1)^n \left[\frac{n-1 - n-1}{n^2-1} \right] \\
&= \frac{(-1)^n - 2}{n^2-1} = \frac{(-1)^{n+1} 2}{n^2-1} \quad \text{for } n=2, 3, \dots
\end{aligned}$$

For $n=1$, the above formula has 0 for denominator, so,

$$\begin{aligned}
a_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin x \cos x \, dx \\
&= \frac{2}{\pi} \times \frac{1}{2} \int_0^{\pi} x 2 \sin x \cos x \, dx \\
&= \frac{1}{\pi} \int_0^{\pi} x \sin 2x \, dx \\
&= \frac{1}{\pi} \left[\frac{-x \cos 2x}{2} + \frac{\sin 2x}{4} \right]_0^{\pi} \\
&= \frac{1}{\pi} \left[x - \frac{\pi \cos 2\pi}{2} \right] \\
&= -\frac{1}{2}
\end{aligned}$$

$$\begin{aligned}
b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} \, dx \\
&= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin x \sin nx \, dx \\
&= 0
\end{aligned}$$

so, from (i),

$$f(x) = 1 - \frac{1}{2} \cos x + \sum_{n=2}^{\infty} \left[\frac{2(-1)^{n+1}}{n^2-1} \cos nx + 0 \right]$$

$$= 1 - \frac{1}{2} \cos x + 2 \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^2-1} \cos nx$$

(v) $f(x) = x^2$; $0 < x < 2\pi$

Let the Fourier series of given function $f(x) = x^2$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (1)}$$

where, $a_0 = \frac{1}{2L} \int_0^{2L} f(x) dx$

$$= \frac{1}{2\pi} \int_0^{2\pi} x^2 dx$$

$$= \frac{1}{2\pi} \left[\frac{x^3}{3} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \times \frac{8\pi^3}{3}$$

$$= \frac{4\pi^2}{3}$$

$$a_n = \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx dx$$

$$= \frac{1}{\pi} \left[\cancel{x^2 \sin nx}^0 + 2x \cos nx - 2 \cancel{\frac{\sin nx}{n^3}}^0 \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \times 4\pi \frac{\cos 2n\pi}{n^2}$$

$$= 4 \times \frac{1}{n^2}$$

$$= \frac{4}{n^2}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx dx \\
 &= \frac{1}{\pi} \left[-\frac{x^2 \cos nx}{n} + 2x \frac{\sin nx}{n^2} + \frac{2 \cos nx}{n^3} \right]_0^{2\pi} \\
 &= \frac{1}{\pi} \left[-4\pi^2 \frac{\cos 2n\pi}{n} + 2 \frac{\cos 2n\pi}{n^3} + 0 - \frac{2}{n^3} \right] \\
 &= \frac{1}{\pi} \left[-4\pi^2 \times \frac{1}{n} + 2 \times \frac{1}{n^3} - \frac{2}{n^3} \right] \\
 &= -4\pi/n
 \end{aligned}$$

So, from (1),

$$\begin{aligned}
 f(x) &= \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right] \\
 &= \frac{4}{3} \pi^2 + 4 \sum_{n=1}^{\infty} \left[\frac{1}{n^2} \cos nx - \frac{\pi}{n} \sin nx \right]
 \end{aligned}$$

(vi) $f(x) = e^{-x}$; $0 < x < 2\pi$

Let the Fourier series of given function $f(x) = e^{-x}$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx] \quad \text{--- (1)}$$

where, $a_0 = \frac{1}{2L} \int_0^{2L} f(x) dx$

$$\begin{aligned}
 &= \frac{1}{2\pi} \int_0^{2\pi} e^{-x} dx \\
 &= \frac{1}{2\pi} [-e^{-x}]_0^{2\pi} \\
 &= \frac{1}{2\pi} (-e^{-2\pi} + 1) \\
 &= \frac{1}{2\pi} (1 - e^{-2\pi})
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cos nx dx \\
 &= \frac{1}{\pi} \left[\frac{-\cos nx e^{-x} + n \sin nx e^{-x}}{1+n^2} \right]_0^{2\pi} \\
 &= \frac{1}{\pi} \left[\frac{-e^{-2\pi} \cos 2n\pi + 1}{1+n^2} \right] \\
 &= \frac{1 - e^{-2\pi}}{\pi (1+n^2)}
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin nx dx \\
 &= \frac{1}{\pi} \left[\frac{-\sin nx e^{-x} - n \cos nx e^{-x}}{1+n^2} \right]_0^{2\pi} \\
 &= \frac{1}{\pi} \left[\frac{-n \cos 2n\pi \cdot e^{-2\pi} + n e^0 \cos 0}{1+n^2} \right] \\
 &= \frac{n (1 - e^{-2\pi})}{\pi (1+n^2)}
 \end{aligned}$$

∴, from (i),

$$\begin{aligned}
 f(x) &= \frac{1}{2\pi} (1 - e^{-2\pi}) + \sum_{n=1}^{\infty} \left[\frac{1 - e^{-2\pi}}{\pi (1+n^2)} \cos nx + \frac{n (1 - e^{-2\pi})}{\pi (1+n^2)} \sin nx \right] \\
 &= \frac{1 - e^{-2\pi}}{2\pi} + \frac{1 - e^{-2\pi}}{\pi} \sum_{n=1}^{\infty} \left[\frac{\cos nx + n \sin nx}{1+n^2} \right]
 \end{aligned}$$

(vii) $f(x) = x - x^2$; $-1 < x < 1$

Let the Fourier series of given function $f(x) = x - x^2$ with period $2L = 2$ (i.e. $L = 1$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$$

$$= a_0 + \sum_{n=1}^{\infty} [a_n \cos n\pi x + b_n \sin n\pi x] \text{--- ①}$$

$$\begin{aligned} \text{where, } a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\ &= \frac{1}{2} \int_{-1}^1 (x - x^2) dx \\ &= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^1 \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{3} - \frac{1}{2} + \frac{1}{3} \right] \\ &= -\frac{1}{3} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ &= \int_{-1}^1 (x - x^2) \cos n\pi x dx \\ &= \int_{-1}^1 x \cos n\pi x dx - \int_{-1}^1 x^2 \cos n\pi x dx \\ &= \left[\frac{x \sin n\pi x}{n\pi} + \frac{\cos n\pi x}{n^2 \pi^2} \right]_{-1}^1 - \left[\frac{x^2 \sin n\pi x}{n\pi} + \frac{2x \cos n\pi x}{n^2 \pi^2} - \frac{2 \sin n\pi x}{n^3 \pi^3} \right]_{-1}^1 \\ &= 0 - \frac{2(-1)^n}{n^2 \pi^2} - \frac{2(-1)^n}{n^2 \pi^2} \\ &= -\frac{4(-1)^n}{n^2 \pi^2} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \\ &= \int_{-1}^1 (x - x^2) \sin n\pi x dx \\ &= \int_{-1}^1 x \sin n\pi x dx - \int_{-1}^1 x^2 \sin n\pi x dx \\ &= 2 \int_0^1 x \sin n\pi x dx - 0 \\ &= 2 \left[-\frac{x \cos n\pi x}{n\pi} + \frac{\sin n\pi x}{n^2 \pi^2} \right]_0^1 \\ &= -\frac{2(-1)^n}{n\pi} \end{aligned}$$

So, from ①,

$$f(x) = -\frac{1}{3} + \sum_{n=1}^{\infty} \left[-\frac{4}{n^2\pi^2} (-1)^n \cos n\pi x - \frac{2}{n\pi} (-1)^n \sin n\pi x \right]$$

$$= -\frac{1}{3} - \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{2}{n^2\pi} (-1)^n \cos n\pi x + \frac{(-1)^n}{n} \sin n\pi x \right]$$

(viii) $f(x) = 2-x$, $0 < x < 2$ and hence deduce: $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}$

Let the Fourier series of given function $f(x) = 2-x$ with period $2L = 2$ (i.e. $L = 1$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} [a_n \cos n\pi x + b_n \sin n\pi x] \quad \text{--- (i)}$$

$$\begin{aligned} \text{where, } a_0 &= \frac{1}{2L} \int_0^{2L} f(x) dx \\ &= \frac{1}{2} \int_0^2 (2-x) dx \\ &= \frac{1}{2} \left[2x - \frac{x^2}{2} \right]_0^2 \\ &= \frac{1}{2} (4 - 2 - 0) \\ &= 1 \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx \\ &= \int_0^2 (2-x) \cos n\pi x dx \\ &= 2 \int_0^2 \cos n\pi x dx - \int_0^2 x \cos n\pi x dx \\ &= 2 \left[\frac{\sin n\pi x}{n\pi} \right]_0^2 - \left[\frac{x \sin n\pi x}{n\pi} + \frac{\cos n\pi x}{n^2\pi^2} \right]_0^2 \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx \\ &= \int_0^2 (2-x) \sin n\pi x dx \\ &= 2 \int_0^2 \sin n\pi x dx - \int_0^2 x \sin n\pi x dx \end{aligned}$$

$$\begin{aligned}
 &= 2 \left[\frac{-\cos n\pi x}{n\pi} \right]_0^1 - \left[\frac{x \cos n\pi x + \sin n\pi x}{n\pi} \right]_0^1 \\
 &= 0 + \frac{2}{n\pi} - 0 \\
 &= \frac{2}{n\pi}
 \end{aligned}$$

∴, from ①,

$$f(x) = 1 + \sum_{n=1}^{\infty} \left[0 + \frac{2}{n\pi} \sin n\pi x \right]$$

$$\text{i.e. } 2-x = 1 + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi x$$

$$\therefore 1-x = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi x \quad \text{--- (ii)}$$

Again,

Put $x = \frac{1}{2}$ in (ii), we get

$$1 - \frac{1}{2} = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{2}$$

$$\therefore \frac{1}{2} \times \frac{\pi}{2} = \sin \frac{\pi}{2} + \frac{1}{2} \sin \pi + \frac{1}{3} \sin \frac{3\pi}{2} + \frac{1}{4} \sin 2\pi + \frac{1}{5} \sin \frac{5\pi}{2} + \dots$$

$$\therefore \frac{\pi}{4} = 1 + 0 - \frac{1}{3} + 0 + \frac{1}{5} + \dots$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \quad \underline{\text{proved}}$$

(ix) $f(x) = \begin{cases} 1 & \text{for } -\pi < x < 0 \\ 0 & \text{for } 0 < x < \pi \end{cases}$ and show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

Let the Fourier series of given function $f(x)$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (i)}$$

$$\begin{aligned}
 \text{where, } a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{2\pi} \int_0^{\pi} f(x) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^0 dx + 0 \\
 &= \frac{1}{2\pi} [x]_{-\pi}^0 \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx dx \\
 &= \frac{1}{\pi} \int_{-\pi}^0 \cos nx dx + 0 \\
 &= \frac{1}{\pi} \left[\frac{1}{n} \sin nx \right]_{-\pi}^0 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_{-\pi}^0 f(x) \sin nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx dx \\
 &= \frac{1}{\pi} \int_{-\pi}^0 \sin nx dx + 0 \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} \cos nx \right]_{-\pi}^0 \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} + \frac{1}{n} (-1)^n \right] \\
 &= \frac{1}{n\pi} [(-1)^n - 1]
 \end{aligned}$$

So, from (1),

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left[0 + \frac{1}{n\pi} [(-1)^n - 1] \sin nx \right]$$

$$= \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} [(-1)^n - 1] \sin nx$$

$$(x) \quad f(x) = \begin{cases} k & \text{for } 0 < x < \pi \\ 0 & \text{for } \pi < x < 2\pi \end{cases}$$

Let the Fourier series of given function $f(x)$ with period

$2L = 2\pi$ (i.e. $L = \pi$) in

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$$

$$= a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx] \quad \text{--- (1)}$$

where, $a_0 = \frac{1}{2L} \int_0^{2L} f(x) dx$

$$= \frac{1}{2\pi} \int_0^{\pi} f(x) dx + \frac{1}{2\pi} \int_{\pi}^{2\pi} f(x) dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} k dx + 0$$

$$= \frac{k}{2\pi} [x]_0^{\pi}$$

$$= \frac{k}{2}$$

$$a_n = \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx dx + \frac{1}{\pi} \int_{\pi}^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} k \cos nx dx + 0$$

$$= \frac{k}{\pi} \left[\frac{1}{n} \sin nx \right]_0^{\pi}$$

$$= 0$$

$$b_n = \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx$$

$$= \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx dx + \frac{1}{\pi} \int_{\pi}^{2\pi} f(x) \sin nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} k \sin nx dx + 0$$

$$= \frac{k}{\pi} \left[-\frac{1}{n} \cos nx \right]_0^{\pi}$$

$$= \frac{k}{\pi} \left[-\frac{1}{n} (-1)^n + \frac{1}{n} \right]$$

$$= \frac{k}{n\pi} (1 - (-1)^n)$$

So, from (1),

$$f(x) = \frac{k}{2} + \sum_{n=1}^{\infty} \left[0 + \frac{k}{n\pi} [1 - (-1)^n] \sin nx \right]$$

$$= \frac{k}{2} + \frac{k}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} [1 - (-1)^n] \sin nx$$

$$(x) f(x) = \begin{cases} x & \text{for } -\pi/2 < x < \pi/2 \\ \pi - x & \text{for } \pi/2 < x < 3\pi/2 \end{cases}$$

Let the Fourier series of given function $f(x)$ with period $2L = 2\pi$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$$

$$= a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx] \quad \text{--- (1)}$$

where, $a_0 = \frac{1}{2L} \int_{-L}^{L} f(x) dx$

$$= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} f(x) dx + \frac{1}{2\pi} \int_{\pi/2}^{3\pi/2} f(x) dx$$

$$= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} x dx + \frac{1}{2\pi} \int_{\pi/2}^{3\pi/2} (\pi - x) dx$$

$$= \frac{1}{2\pi} \left[\frac{x^2}{2} \right]_{-\pi/2}^{\pi/2} + \frac{1}{2\pi} \left[\pi x - \frac{x^2}{2} \right]_{\pi/2}^{3\pi/2}$$

$$= \frac{1}{2\pi} \times 0 + \frac{1}{2\pi} \left[\frac{3\pi^2}{2} - \frac{9\pi^2}{8} - \frac{\pi^2}{2} + \frac{\pi^2}{8} \right]$$

$$= \frac{1}{2\pi} \times 0$$

$$= 0$$

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \cos nx dx + \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} x \cos nx dx + \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} (\pi - x) \cos nx dx$$

$$= \frac{1}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_{-\pi/2}^{\pi/2} + \frac{1}{\pi} \left[\frac{(\pi - x) \sin nx}{n} - \frac{\cos nx}{n^2} \right]_{\pi/2}^{3\pi/2}$$

$$= \frac{1}{\pi} \left[\frac{\pi}{2n} \sin \frac{n\pi}{2} + \frac{1}{n^2} \cos \frac{n\pi}{2} + \frac{\pi}{2n} \sin \frac{-n\pi}{2} - \frac{1}{n^2} \cos \frac{-n\pi}{2} \right]$$

$$+ \frac{1}{\pi} \left[-\frac{\pi}{2n} \sin \frac{3n\pi}{2} - \frac{1}{n^2} \cos \frac{3n\pi}{2} - \frac{\pi}{2n} \sin \frac{n\pi}{2} + \frac{1}{n^2} \cos \frac{n\pi}{2} \right]$$

$$= 0 + \frac{1}{\pi} \times \left(-\frac{\pi}{2n} \right) [\sin \frac{3n\pi}{2} + \sin \frac{n\pi}{2}] + \frac{1}{\pi} \times \left(-\frac{1}{n^2} \right) [\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2}]$$

$$= -\frac{1}{2n} \times 2 \sin n\pi \cos \frac{n\pi}{2} - \frac{1}{n^2} \pi \times [-2 \sin n\pi \sin \frac{n\pi}{2}]$$

$$= -\frac{1}{n} \sin n\pi \cos \frac{n\pi}{2} + \frac{2}{n^2} \pi \sin n\pi \sin \frac{n\pi}{2}$$

$$\begin{aligned}
 b_n &= \frac{1}{2L} \int_{-\pi/2}^{3\pi/2} f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \sin nx dx + \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} f(x) \sin nx dx \\
 &= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} x \sin nx dx + \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} (\pi - x) \sin nx dx \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} x \cos nx + \frac{1}{n^2} \sin nx \right]_{-\pi/2}^{\pi/2} + \frac{1}{\pi} \left[-(\pi - x) \cdot \frac{1}{n} \cos nx \right. \\
 &\quad \left. - \frac{1}{n^2} \sin nx \right]_{\pi/2}^{3\pi/2} \\
 &= \frac{1}{\pi} \left[-\frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} - \frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} \right] \\
 &\quad + \frac{1}{\pi} \left[\frac{\pi}{2n} \cos \frac{3n\pi}{2} - \frac{1}{n^2} \sin \frac{3n\pi}{2} + \frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} \right] \\
 &= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos \frac{n\pi}{2} + \frac{2}{n^2} \sin \frac{n\pi}{2} + \frac{\pi}{2n} \cos \frac{3n\pi}{2} - \frac{1}{n^2} \sin \frac{3n\pi}{2} \right. \\
 &\quad \left. + \frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} \right] \\
 &= \frac{1}{\pi} \left[-\frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{3}{n^2} \sin \frac{n\pi}{2} + \frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} \right] \\
 &= \frac{1}{\pi} \left[\frac{4}{n^2} \sin \frac{n\pi}{2} \right] \\
 &= \frac{4}{n^2} \sin \frac{n\pi}{2}
 \end{aligned}$$

So, from (i).

$$\begin{aligned}
 f(x) &= 0 + \sum_{n=1}^{\infty} \left[\left(-\frac{1}{n} \sin n\pi \cos \frac{n\pi}{2} + \frac{2}{n^2} \pi \sin n\pi \sin \frac{n\pi}{2} \right) \cos nx + \right. \\
 &\quad \left. \frac{4}{n^2} \pi \sin \frac{n\pi}{2} \sin nx \right] \\
 &= \sum_{n=1}^{\infty} \left[\left(-\frac{1}{n} \sin n\pi \cos \frac{n\pi}{2} + \frac{2}{n^2} \pi \sin n\pi \sin \frac{n\pi}{2} \right) \cos nx + \frac{4}{n^2} \pi \sin \frac{n\pi}{2} \right. \\
 &\quad \left. \sin nx \right]
 \end{aligned}$$

$$(xii) \quad f(x) = \begin{cases} 0 & \text{for } 0 < x < L \\ 1 & \text{for } L < x < 2L \end{cases}$$

Let the Fourier series of given function $f(x)$ with period $2L$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \quad \text{--- (1)}$$

$$\text{where, } a_0 = \frac{1}{2L} \int_0^{2L} f(x) dx$$

$$\begin{aligned}
 &= \frac{1}{2L} \int_0^L f(x) dx + \frac{1}{2L} \int_L^{2L} f(x) dx \\
 &= \frac{1}{2L} \int_0^L 0 dx + \frac{1}{2L} \int_L^{2L} 1 dx \\
 &= \frac{1}{2L} [x]_L^{2L} = \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx \\
 &= \frac{1}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx + \frac{1}{L} \int_L^{2L} f(x) \cos \frac{n\pi x}{L} dx \\
 &= 0 + \frac{1}{L} \int_L^{2L} \cos \frac{n\pi x}{L} dx \\
 &= \frac{1}{L} \left[\frac{L}{n\pi} \sin \frac{n\pi x}{L} \right]_L^{2L} \\
 &= \frac{1}{L} \left[\frac{L}{n\pi} \sin 2n\pi - \frac{L}{n\pi} \sin n\pi \right] \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx + \frac{1}{L} \int_L^{2L} f(x) \sin \frac{n\pi x}{L} dx \\
 &= 0 + \frac{1}{L} \int_L^{2L} \sin \frac{n\pi x}{L} dx \\
 &= \frac{1}{L} \left[-\frac{L}{n\pi} \cos \frac{n\pi x}{L} \right]_L^{2L} \\
 &= \frac{1}{L} \times -\frac{L}{n\pi} [\cos 2n\pi - \cos n\pi] \\
 &= -\frac{1}{n\pi} [1 - (-1)^n] \\
 &= \begin{cases} 0 & \text{for even } n \\ -\frac{2}{n\pi} & \text{for odd } n \end{cases}
 \end{aligned}$$

∴, from (i),

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} (0 + b_n \sin \frac{n\pi x}{L})$$

$$= \frac{1}{2} + \sum_{n=1,3,5,\dots}^{\infty} \left(-\frac{2}{n\pi} \sin \frac{n\pi x}{L} \right)$$

$$= \frac{1}{2} - \frac{2}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{L}$$

Fourier Sine and Cosine Series:

3 Find the Fourier sine and cosine series of $f(x)=x$; $0 < x < L$

✓ Let the Fourier sine series of given function $f(x)=x$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} \quad \text{--- (i)}$$

$$\begin{aligned}
 \text{where, } b_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \int_0^L x \sin \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \left[-\frac{L}{n\pi} x \cos \frac{n\pi x}{L} + \frac{L^2}{n^2\pi^2} \sin \frac{n\pi x}{L} \right]_0^L \\
 &= \frac{2}{L} \left[-\frac{L}{n\pi} \times L \cos n\pi + 0 \right] \\
 &= -\frac{2L}{n\pi} (-1)^n
 \end{aligned}$$

So, from (i),

$$\begin{aligned}
 f(x) &= \sum_{n=1}^{\infty} -\frac{2L}{n\pi} (-1)^n \sin \frac{n\pi x}{L} \\
 &= -\frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin \frac{n\pi x}{L}
 \end{aligned}$$

Again,

Let the Fourier cosine series of given function $f(x) = x$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} \quad \text{--- (ii)}$$

$$\begin{aligned}
 \text{where, } a_0 &= \frac{1}{L} \int_0^L f(x) dx \\
 &= \frac{1}{L} \int_0^L x dx \\
 &= \frac{1}{L} \left[\frac{x^2}{2} \right]_0^L \\
 &= \frac{L}{2}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \int_0^L x \cos \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \left[\frac{L}{n\pi} x \sin \frac{n\pi x}{L} + \frac{L^2}{n^2\pi^2} \cos \frac{n\pi x}{L} \right]_0^L \\
 &= \frac{2}{L} \times \frac{L^2}{n^2\pi^2} [(-1)^n - 1] \\
 &= \frac{2L}{n^2\pi^2} [(-1)^n - 1] \\
 &= \begin{cases} 0 & \text{for even } n \\ -\frac{4L}{n^2\pi^2} & \text{for odd } n \end{cases}
 \end{aligned}$$

So, from (i),

$$f(x) = L/2 + \sum_{n=1,3,5,\dots}^{\infty} -4L/n^2\pi^2 \cos \frac{n\pi x}{L}$$

$$= L/2 - 4L/\pi^2 \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{L}$$

4. Find the Fourier sine and cosine series of $f(x) = \pi - x$; $0 < x < \pi$

Let the Fourier sine series of given function $f(x) = \pi - x$ defined on $[0, L] = [0, \pi]$ (i.e. $L = \pi$) is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$= \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (i)}$$

$$\text{where, } b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi - x) \sin nx dx$$

$$= \frac{2}{\pi} \left[-\frac{1}{n} (\pi - x) \cos nx - \frac{1}{n^2} \sin nx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \times \frac{\pi}{n} \times 1$$

$$= \frac{2}{n}$$

So, from (i),

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n} \sin nx$$

$$= 2 \sum_{n=1}^{\infty} \frac{1}{n} \sin x$$

Again,

Let the Fourier cosine series of given function $f(x) = \pi - x$ defined on $[0, L] = [0, \pi]$ (i.e. $L = \pi$) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \cosh x \quad \text{--- (ii)}$$

where, $a_0 = \frac{1}{L} \int_0^L f(x) dx$

$$= \frac{1}{\pi} \int_0^{\pi} (\pi - x) dx$$

$$= \frac{1}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left(\pi^2 - \frac{\pi^2}{2} - 0 \right)$$

$$= \frac{1}{\pi} \times \frac{\pi^2}{2}$$

$$= \frac{\pi}{2}$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi - x) \cosh nx dx$$

$$= \frac{2}{\pi} \left[\frac{1}{n} (\pi - x) \sinh nx - \frac{1}{n^2} \cosh nx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-\frac{1}{n^2} (-1)^n + \frac{1}{n^2} \right]$$

$$= \frac{2}{n^2} \pi [1 - (-1)^n]$$

$$= \begin{cases} 0 & \text{for even } n \\ \frac{4}{n^2} \pi & \text{for odd } n \end{cases}$$

So, from (ii),

$$f(x) = \frac{\pi}{2} + \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{n^2} \pi \cosh nx$$

$$= \frac{\pi}{2} + \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cosh nx$$

5. Find the half range sine series for $f(x) = \pi x - x^2$ in the interval $0 < x < \pi$

✓ The half range sine series for $f(x) = \pi x - x^2$ defined on $[0, L] = [0, \pi]$ (i.e. $L = \pi$) can be written as

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$= \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

where, $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi x - x^2) \sin nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \pi x \sin nx dx - \frac{2}{\pi} \int_0^{\pi} x^2 \sin nx dx$$

$$= 2 \left[-\frac{1}{n} x \cos nx + \frac{1}{n^2} \sin nx \right]_0^{\pi} - \frac{2}{\pi} \left[-\frac{1}{n} x^2 \cos nx + \frac{2}{n^2} x \sin nx + \frac{2}{n^3} \cos nx \right]_0^{\pi}$$

$$= 2 \times -\frac{\pi}{n} (-1)^n - \frac{2}{\pi} \left[-\frac{\pi^2}{n} (-1)^n + \frac{2}{n^3} (-1)^n - \frac{2}{n^3} \right]$$

$$= -\frac{2\pi}{n} (-1)^n + \frac{2\pi}{n} (-1)^n - \frac{4}{\pi n^3} (-1)^n + \frac{4}{\pi n^3}$$

$$= \frac{4}{\pi n^3} [1 - (-1)^n]$$

$$= \begin{cases} 0 & \text{for even } n \\ \frac{8}{\pi n^3} & \text{for odd } n \end{cases}$$

So, from (1),

$$f(x) = \sum_{n=1,3,5,\dots}^{\infty} \frac{8}{\pi n^3} \sin nx$$

$$= \frac{8}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^3} \sin nx$$

6. Find the Fourier sine and cosine series of $f(x) = e^x$, $0 < x < L$.
 Let the Fourier sine series of given function $f(x) = e^x$ defined on $[0, L]$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} \quad \text{--- (1)}$$

where, $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$

$$= \frac{2}{L} \int_0^L e^x \sin \frac{n\pi x}{L} dx$$

$$\begin{aligned}
 &= \frac{2}{L} \left[\frac{e^x}{1 + n^2 \pi^2 / L^2} \left(\sin \frac{n\pi x}{L} - \frac{n\pi}{L} \cos \frac{n\pi x}{L} \right) \right]_0^L \\
 &= \frac{2}{L} \left[\frac{e^L}{1 + n^2 \pi^2 / L^2} [0 - n\pi(-1)^n] - \frac{1}{1 + n^2 \pi^2 / L^2} [0 - \frac{n\pi}{L}] \right] \\
 &= \frac{2}{L} \cdot \frac{e^L}{1 + n^2 \pi^2 / L^2} \cdot (-1)^n n\pi + \frac{2}{L} \cdot \frac{1}{1 + n^2 \pi^2 / L^2} \cdot \frac{n\pi}{L} \\
 &= \frac{2n\pi}{L^2 + n^2 \pi^2} (1 - (-1)^n e^L)
 \end{aligned}$$

So, from (i),

$$f(x) = \sum_{n=1}^{\infty} \frac{2n\pi}{L^2 + n^2 \pi^2} [1 - (-1)^n e^L] \sin \frac{n\pi x}{L}$$

Again,

Let the Fourier cosine series of given function $f(x) = e^x$ defined on $[0, L]$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} \quad \text{--- (ii)}$$

$$\begin{aligned}
 \text{where, } a_0 &= \frac{1}{L} \int_0^L f(x) dx \\
 &= \frac{1}{L} \int_0^L e^x dx \\
 &= \frac{1}{L} [e^x]_0^L \\
 &= \frac{1}{L} (e^L - 1)
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \int_0^L e^x \cos \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \left[\frac{e^x}{1 + n^2 \pi^2 / L^2} \left(\cos \frac{n\pi x}{L} + \frac{n\pi}{L} \sin \frac{n\pi x}{L} \right) \right]_0^L \\
 &= \frac{2}{L} \times \frac{L^2}{(L^2 + n^2 \pi^2)} [e^L (-1)^n - 1.1] \\
 &= \frac{2L}{L^2 + n^2 \pi^2} [e^L (-1)^n - 1]
 \end{aligned}$$

So, from (ii),

$$f(x) = \frac{1}{L}(e^L - 1) + \sum_{n=1}^{\infty} \frac{2L}{L^2 + n^2\pi^2} [e^L(-1)^n - 1] \cos \frac{n\pi x}{L}$$

Partial Differential Equations:

7. Find the general solution of $2u_x + 2u_y - u = 0$.

✓ Comparing the given equation with $au_x + bu_y - c = 0$, we get
 $a = 2, \quad b = 2, \quad c = u$

Then the characteristics eqⁿ are

$$\frac{dx}{a} = \frac{dy}{b} = \frac{dz}{c}$$

$$\frac{dx}{2} = \frac{dy}{2} = \frac{dz}{u}$$

$$\therefore \frac{dx}{2} = \frac{dy}{2} = \frac{dz}{u}$$

Taking first two ratios,
 $\frac{dx}{2} = \frac{dy}{2}$

$$\therefore dx = dy$$

$$\therefore x = y + c_1$$

$$\text{i.e. } c_1 = x - y$$

Taking last two ratios,
 $\frac{du}{u} = \frac{dy}{2}$

$$\therefore \ln u = \frac{y}{2} + c_2$$

$$\therefore u = e^{\frac{y}{2}} e^{c_2} = c_2 e^{\frac{y}{2}}$$

$$\text{i.e. } c_2 = \frac{u}{e^{\frac{y}{2}}}$$

Then the general solution of given eqⁿ is

$$c_2 = F(c_1)$$

$$\therefore \frac{u}{e^{\frac{y}{2}}} = F(x - y)$$

$$\therefore u = e^{\frac{y}{2}} F(x - y)$$

8. Find the solution of $u_x + u_y = u$ given that $u(x, 0) = u_0(x) = 2$
 comparing the given equation with $au_x + by_y - c = 0$, we get,
 $a = 1, b = 1, c = u$

Then the characteristic eqⁿ are

$$dx/a = dy/b = du/c$$

$$dx/1 = dy/1 = du/c$$

$$\therefore \frac{dx}{1} = \frac{dy}{1}$$

$$\text{and } \frac{du}{u} = \frac{dy}{1}$$

$$\therefore dx = dy$$

$$\therefore \ln u = y + c_2$$

$$\therefore x = y + c_1$$

$$\therefore u = e^y \cdot c_2$$

$$\text{i.e. } c_1 = x - y$$

$$\text{i.e. } c_2 = \frac{u}{e^y}$$

Then, the general solution of given eqⁿ is

$$c_2 = F(c_1)$$

$$\therefore \frac{u}{e^y} = F(x - y)$$

$$\therefore u = e^y F(x - y)$$

$$\text{i.e. } u(x, y) = e^y F(x - y)$$

$$\therefore u(x, 0) = e^0 F(x - 0)$$

$$\therefore 2 = F(x)$$

Hence, the solution is

$$u = e^y F(x - y)$$

$$\text{i.e. } u = e^y \cdot 2$$

$$\therefore u = 2e^y$$

~~Final~~
11/03